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POWER TOOL INCLUDING SECONDARY ENERGY STORAGE

Abstract

Examples provide a power tool including a power input interface electrically connectable to a power source; a motor; a DC-DC power converter electrically connected to the motor; a secondary energy storage device electrically connected to the DC-DC power converter; and a controller electrically connected to the DC-DC power converter, the controller configured to control the DC-DC power converter to route a braking current generated by the motor to the secondary energy storage device during a regenerative braking operation.

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Background/Summary

FIELD

[0001] Examples described herein relate to power tools.

SUMMARY

[0002] Power tools described herein include a power input interface electrically connectable to a power source; a motor; a DC-DC power converter electrically connected to the motor; a secondary energy storage device electrically connected to the DC-DC power converter; and a controller electrically connected to the DC-DC power converter, the controller configured to control the DC-DC power converter to route a braking current generated by the motor to the secondary energy storage device during a regenerative braking operation.

[0003] In some aspects, the DC-DC power converter is configured to regulate power provided to the secondary energy storage device from the motor during the regenerative braking operation.

[0004] In some aspects, the DC-DC power converter is a bidirectional DC-DC power converter configured to regulate power discharged from the secondary energy storage device.

[0005] In some aspects, the power tool further includes a power switching network electrically connected to the motor and the DC-DC power converter, wherein the controller is configured to control the power switching network to perform the regenerative braking operation.

[0006] In some aspects, the controller is further configured to selectively provide energy stored in the secondary energy storage device to one of the motor or to the power source.

[0007] In some aspects, the controller is configured to selectively provide the energy stored in the secondary energy storage device to the motor as a motor plugging current during a braking operation of the motor.

[0008] In some aspects, the controller is configured to selectively provide the energy stored in the secondary energy storage device to the motor as supplementary power for driving the motor.

[0009] In some aspects, the controller is configured to provide power from the power source and the supplementary power from the secondary energy storage device to the motor at startup.

[0010] In some aspects, the controller is configured to provide the supplementary power in response to determination that a voltage of the power source is below a threshold during a driving operation of the motor.

[0011] In some aspects, the controller is configured to provide the supplementary power in response to determination that a power demand of the motor is greater than a threshold.

[0012] In some aspects, the power tool further includes a user input interface; wherein the controller is further configured to receive user input via the user input interface indicating that a boost operation is requested, and selectively provide the energy stored in the secondary energy storage device to the motor as supplementary power for driving the motor in response to determining that the boost operation is requested.

[0013] In some aspects, the controller is configured to selectively provide the energy stored in the secondary energy storage device to the power source in response to determining that the motor is in an idle state.

[0014] In some aspects, the secondary energy storage device includes at least one selected from the group consisting of a capacitor, a supercapacitor, a battery cell, and a power tool battery pack.

[0015] In some aspects, the power source is a removable battery pack.

[0016] Power tools described herein include: a battery pack interface electrically connectable to a battery pack; a motor; a DC-DC power converter electrically connected to the motor; a secondary energy storage device electrically connected to the DC-DC power converter; and a controller electrically connected to the DC-DC power converter, the controller configured to control DC-DC power converter to route a braking current generated by the motor to the secondary energy storage

device during a regenerative braking operation.

[0017] In some aspects, the controller is further configured to responsive to determining that a first operating condition of the power tool is met, selectively provide energy stored in the secondary energy storage device to the motor, responsive to determining that the first operating condition of the power tool is not met and that a second operating condition of the power tool is met, selectively provide the energy stored in the secondary energy storage device to the power tool battery pack, and responsive to determining that the first operating condition and the second operating condition are not met, selectively enable the energy stored in the energy storage device to dissipate.

[0018] Methods for operating a power tool including a power input interface electrically connectable to a power source, a motor, a DC-DC power converter, and a secondary energy storage device described herein include: controlling the DC-DC power converter to route a braking current generated by the motor to the secondary energy storage device during a regenerative braking operation.

[0019] In some aspects, the method further includes selectively providing energy stored in the secondary energy storage device to one selected from a group consisting of the motor and the power source.

[0020] In some aspects, the energy stored in the secondary energy storage device to the motor is selectively provided to the motor as a motor plugging current during a braking operation of the motor.

[0021] In some aspects, the energy stored in the secondary energy storage device to the motor is selectively provided to the motor as supplementary power for driving the motor.

[0022] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in application to the details of the configurations and arrangements of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

[0023] Unless the context of their usage unambiguously indicates otherwise, the articles “a,” “an,” and “the” should not be interpreted as meaning “one” or “only one.” Rather these articles should be interpreted as meaning “at least one” or “one or more.” Likewise, when the terms “the” or “said” are used to refer to a noun previously introduced by the indefinite article “a” or “an,” “the” and “said” mean “at least one” or “one or more” unless the usage unambiguously indicates otherwise.

[0024] In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers,” “computing devices,” “controllers,” “processors,” etc., described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

[0025] Relative terminology, such as, for example, “about,” “approximately,” “substantially,” etc., used in connection with a quantity or condition would be understood by those of ordinary skill to

be inclusive of the stated value and has the meaning dictated by the context (e.g., the term includes at least the degree of error associated with the measurement accuracy, tolerances [e.g., manufacturing, assembly, use, etc.] associated with the particular value, etc.). Such terminology should also be considered as disclosing the range defined by the absolute values of the two endpoints. For example, the expression “from about 2 to about 4” also discloses the range “from 2 to 4”. The relative terminology may refer to plus or minus a percentage (e.g., 1%, 5%, 10%) of an indicated value.

[0026] It should be understood that although certain drawings illustrate hardware and software located within particular devices, these depictions are for illustrative purposes only. Functionality described herein as being performed by one component may be performed by multiple components in a distributed manner. Likewise, functionality performed by multiple components may be consolidated and performed by a single component. In some embodiments, the illustrated components may be combined or divided into separate software, firmware and/or hardware. For example, instead of being located within and performed by a single electronic processor, logic and processing may be distributed among multiple electronic processors. Regardless of how they are combined or divided, hardware and software components may be located on the same computing device or may be distributed among different computing devices connected by one or more networks or other suitable communication links. Similarly, a component described as performing particular functionality may also perform additional functionality not described herein. For example, a device or structure that is “configured” in a certain way is configured in at least that way but may also be configured in ways that are not explicitly listed.

[0027] Accordingly, in the claims, if an apparatus, method, or system is claimed, for example, as including a controller, control unit, electronic processor, computing device, logic element, module, memory module, communication channel or network, or other element configured in a certain manner, for example, to perform multiple functions, the claim or claim element should be interpreted as meaning one or more of such elements where any one of the one or more elements is configured as claimed, for example, to make any one or more of the recited multiple functions, such that the one or more elements, as a set, perform the multiple functions collectively.

[0028] Other aspects of the embodiments will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] FIG. 1 illustrates a power tool, according to some examples.

[0030] FIG. 2 schematically illustrates a power tool, according to some examples.

[0031] FIG. 3 schematically illustrates an energy management system for a power tool, according to some examples.

[0032] FIG. 4 illustrates a method for controlling a power tool, according to some examples.

DETAILED DESCRIPTION

[0033] FIG. 1 illustrates a power tool **100** including a brushless direct current (“BLDC”) motor. The power tool **100** is, for example, a brushless hammer drill including a housing **102**. The housing **102** includes a handle portion **104** and motor housing portion **106**. The power tool **100** further includes an output driver **108** (illustrated as a chuck), a trigger **110**, and a power input interface **112**. The power input interface **112** is configured to mechanically and electrically connect to a power source for powering the tool **100**. Although FIG. 1 illustrates a hammer drill, in some embodiments, the components and control techniques described herein are incorporated into other types of power tools including, for example, a circular saw, a jigsaw, a reciprocating saw, a bandsaw, a grinder, a cutoff saw, a tire buffer, a mud mixer, a bandfile, a polisher, a sander, a cutoff

tool, a rotary hammer, a drill-driver, a hammer drill, a right angle drill, an impact driver, an impact wrench, a ratchet, a screwdriver, a crimper, a pipe threader, a pump, a cable cutter, a cable stripper, a rod cutter, a tube cutter, a pipe shear, a knockout tool, a PEX expander, an inflator, a compressor, a sewer drum, a transfer pump, a drain snake, a rivet tool, a heat gun, a grease gun, a caulk gun, a chain hoist, a track saw, a miter saw, a table saw, a multi-tool, a router, a planer, a vacuum, a fan, a blower, etc.

[0034] FIG. 2 illustrates a control system **200** for the power tool **100** that includes a secondary energy storage system. The control system **200** includes a controller **202**. The controller **202** is electrically and/or communicatively connected to a variety of modules or components of the power tool **100**. For example, the illustrated controller **202** is electrically connected to a motor **204**, the power input interface **112**, a trigger switch **208** (connected to the trigger **110**), one or more sensors **212** (also referred to as sensing circuits), one or more indicators **214**, a user input module **216**, a power switching network **220**, a secondary energy storage device **222**, and a DC-DC converter **224**. The controller **202** includes combinations of hardware and software that are operable to, among other things, control the operation of the power tool **100**, monitor the operation of the power tool **100**, activate the one or more indicators **214** (e.g., an LED), etc.

[0035] The power input interface **112** is electrically connectable to a power source **242**. The power source **242** may be, for example, a rechargeable and removable power tool battery pack. While described herein as a battery pack **242**, in some examples, the power source **242** is, for example, a portable power supply having a corded connection to the power input interface **112**, an adapter for converting alternating current to direct current supplied to the power input interface, a power cord providing alternating current power input, or another type of power source. The power input interface **112** includes a combination of mechanical components (e.g., rails, grooves, latches, etc.) and electrical components (e.g., one or more terminals) configured to and operable for interfacing (e.g., mechanically, electrically, and communicatively connecting) with the battery pack **242**. The power input interface **112** supplies power to the power switching network **220** to be switched by the switching FETs to selectively provide power to the motor **204**.

[0036] The controller **202** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **202** and/or the power tool **100**. For example, the controller **202** includes, among other things, a processing unit **226** (e.g., a microprocessor, a microcontroller, an electronic controller, an electronic processor, or another suitable programmable device and may be commonly referred to as an electronic processor), a memory **228**, input units **230**, and output units **232**. The processing unit **226** includes, among other things, a control unit **234**, an arithmetic logic unit (“ALU”) **236**, and a plurality of registers **238**, and is implemented using a known computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit **226**, the memory **228**, the input units **230**, and the output units **232**, as well as the various modules or circuits connected to the controller **202** are connected by one or more control and/or data buses (e.g., common bus **240**). The control and/or data buses are shown generally in FIG. 2 for illustrative purposes. The use of one or more control and/or data buses for the interconnection between and communication among the various modules, circuits, and components would be known to a person skilled in the art in view of the invention described herein.

[0037] The memory **228** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit **226** is connected to the memory **228** and executes software instructions that are capable of being stored in a RAM of the memory **228** (e.g., during execution), a ROM of the memory **228** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc.

Software included in the implementation of the power tool **100** can be stored in the memory **228** of the controller **202**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **202** is configured to retrieve from the memory **228** and execute, among other things, instructions related to the control processes and methods described herein. In other constructions, the controller **202** includes additional, fewer, or different components.

[0038] In the example illustrated, the motor **204** is a three-phase brushless motor. In some example, the motor **204** may have more of fewer number of phases. The power switching network **220** includes, for example, an inverter having a plurality of switching FETs for driving the motor **204**. In one example, the power switching network **220** is formed as an inverter bridge including three high-side switching FETs connected between a positive terminal of a power supply and motor terminals and three low-side switching FETs connected between the motor terminals and a negative terminal of the power supply or ground. In other examples, the motor **204** is a DC motor or a brushed motor and the power switching network **220** is modified accordingly. In some embodiments, the sensors **212** measure or detect a current in the power switching network **220** (e.g., a current of one or more phases of the inverter).

[0039] The trigger switch **208** provides information relating to the trigger **110** to the controller **202**. In some examples, the trigger **110** is an ON/OFF power switch. In these examples, the trigger switch **208** may be connected between the power input interface **112** and power switching network **220** and may include a relay that is closed when the trigger **110** is turned ON and opened with the trigger **110** is turned OFF. In other examples, the trigger **110** may include a variable speed trigger. In these examples, the trigger switch **208** may include a wiper in addition to or in place of a relay. As the variable speed trigger is depressed, contacts of the variable speed trigger move along the wiper of the trigger switch **208** thereby providing the desired speed information to the controller **202**.

[0040] The sensors **212** include one or more current sensors, one or more speed sensors, one or more Hall Effect sensors, one or more temperature sensors, etc. The indicators **214** include, for example, one or more light-emitting diodes (“LEDs”). The indicators **214** can be configured to display conditions of, or information associated with, the power tool **100**. For example, the indicators **214** are configured to indicate measured electrical characteristics of the power tool **100**, the status of the power tool, the status of the secondary energy storage device **222**, etc. The user input module **216** is operably coupled to the controller **202** to, for example, select a forward mode of operation or a reverse mode of operation, a torque and/or speed setting for the power tool **100** (e.g., using torque and/or speed switches), etc. In some embodiments, the user input module **216** includes a combination of digital and analog input or output devices required to achieve a desired level of operation for the power tool **100**, such as one or more knobs, one or more dials, one or more switches, one or more buttons, etc.

[0041] Referring to FIGS. **2** and **3**, the secondary energy storage device **222** may include, for example, a capacitor, a supercapacitor, a battery pack (e.g., a secondary battery pack in addition to the battery pack **242**), and/or the like. In some examples, the controller **202** may be directly connected to the secondary energy storage device **222**. In other examples, the controller **202** may determine a state of the energy storage device using the sensors **212**. A DC-DC power converter **224** is electrically connected to the secondary energy storage device **222**, the power switching network **220**, and the controller **202**. In some examples, the DC-DC power converter **224** is connected between the power switching network **220** (e.g., an inverter) and the secondary energy storage device **222**. In other examples, DC-DC power converter **224** may be connected directly to the motor **204** or connected to the motor via another intermediary power converter (e.g., a rectifier). In some examples, the DC-DC power converter **224** may also be connected to the power source **242** through the power input interface **112**. Although FIG. **3** illustrates the main power source **242** as a battery and the secondary energy storage device **222** as a super-capacitor, the power

source **242** and/or the secondary energy storage device **222** may take other forms as described above.

[0042] FIG. **3** illustrates an energy management system of the power tool **100**. The power switching network **220** is connected between a DC bus **312** and the motor **204**. The power switching network **220** may include a bidirectional circuit that selectively provides power from the DC bus **312** to the motor **204** (e.g., when driving the motor) and from the motor **204** to the DC bus **312** (e.g., when braking the motor). The DC bus **312** is also connected to the power source **242**, the secondary energy storage device **222**, and the DC-DC power converter **224**. The DC-DC power converter **224** may be a bidirectional DC-DC power converter **224** that provides DC power from the secondary energy storage device **222** to the DC bus **312** and the DC bus **312** to the secondary energy storage device **222**.

[0043] The controller **202** is configured to monitor the states of the power source **242**, the secondary energy storage device **222**, and the motor. For example, the controller **202** may monitor the voltage level (e.g., state of charge), the power draw, the temperature, error states, and the like of the power source **242** and the secondary energy storage device **222**. The controller **202** may also monitor the power demand (e.g., the load) on the motor, for example, using load sensor, vibration sensors, and the like. The controller **202** provides control signals to the DC-DC power converter **224** to control the power flow between the DC bus **312** and the secondary energy storage device **222**.

[0044] In some example, switches (e.g., FETs) may be arranged on the current path between the DC bus **312** and each of power source **242**, the power switching network **220**, and the DC-DC power converter **224** respectively. The switches selectively connect and disconnect the power source **242**, the power switching network **220**, and the DC-DC converter to the DC bus **312**.

[0045] In some examples, the DC-DC power converter **224** is a bidirectional power converter configured to regulate power (e.g., boost or buck voltage) provided to and discharged from the secondary energy storage device **222** automatically or based on control signals received from the controller **202**. For example, the DC-DC power converter **224** may regulate power discharged from the secondary energy storage device **222** that is provided to the motor **204** (e.g., via the power switching network **220**) and/or the power source **242** (e.g., via the power input interface **112**).

[0046] FIG. **4** illustrates a method **400** for energy management of the power tool **100**. The method **400** may be performed by, for example, the controller **202** of the power tool **100**. The method **400** includes detecting, using the controller **202**, a motor driving event (at block **404**). The motor driving event may include, for example, actuation of a power switch, a variable speed trigger (e.g., the trigger **110**), actuation of a power switch and a speed dial, and/or the like. The controller **202** may detect the motor driving event based on signals from the trigger switch **208** or the sensors **212**.

[0047] The method **400** includes, in response to detecting the motor driving event, driving, using the power switching network **220**, the motor **204** according to the motor driving event (e.g., according to a speed and/or torque associated with the motor driving event) (at block **408**). During operation of the power tool **100**, the controller **202** is configured to control the power switching network **220** to drive the motor **204** in response to, for example, detecting actuation of the trigger **110**. As described above, the power switching network **220** may be formed as an inverter electrically connected between the DC bus **312** and the motor **204**. The controller **202** may provide pulse width modulated (PWM) signals to the FETs of the power switching network **220** to drive the motor **204** according to the motor driving event. Driving the motor **204** causes a motor driving current to flow from, for example, the DC bus **312** to the motor **204** via the power switching network **220**.

[0048] The method **400** also includes detecting, using the controller **202**, a motor braking event (at block **412**). The motor braking event may include, for example, a release of the trigger **110**, actuation of a power switch into an OFF state, a speed reduction according to a speed dial of the tool **100**, a kickback event of the tool **100**, actuation of an emergency stop input included in the

user input module **216**, a torque limit of the motor **204** being reached, and/or the like. The controller **202** may detect the motor driving event based on signals from the trigger switch **208** or the sensors **212**.

[0049] The method **400** includes braking, using the controller **202**, the motor in response to the motor braking event (at block **416**). The controller **202** may brake the motor by disconnecting the power source **242** from the DC bus **312**. The method **400** further includes controlling the DC-DC power converter **224** to provide a current generated by the motor **204** to the secondary energy storage device **222**, the current being generated during motor braking (at block **420**). The controller **202** may control the power switching network **220** to perform a regenerative braking operation of the motor **204** that causes a braking current generated by the motor **204** to be redirected to the secondary energy storage device **222** through the DC-DC power converter. For example, the controller **202** may control a switch arranged between the power source **242** and the power switching network **220** to open such that braking current generated by the motor **204** is routed to the secondary energy storage device **222** via the DC-DC power converter **224**. During the regenerative braking operation, the DC-DC power converter **224** may convert the regenerative braking current received via the power switching network **220** to an appropriate voltage for charging the secondary energy storage device **222**. In some examples, the power switching network **220** may be bypassed (e.g., using a bypass switching arrangement) such that the DC-DC power converter **224** is connected to the motor **204** (that is, motor coils) directly or through a rectifier.

[0050] The method **400** further includes selectively providing, using the controller **202** energy stored in the secondary energy storage device **222** to one of the motor **204** or the power source **242** (at block **424**). Based on state information received from the secondary energy storage device **222**, the battery pack **242**, the motor **204**, and/or other components of the tool **100**, the controller **202** may provide the stored energy in the secondary energy storage device **222** to one of the motor **204** or the power source **242**. For example, the controller **202** may determine a charge state (e.g., a voltage level) of the battery pack **242**, a charge state of the secondary energy storage device **222**, a power demand of the motor **204**, a user input of the tool **100**, and/or one or more operations executed by the motor (e.g., a motor braking operation, a motor driving operation, etc.), and output a control signal to the power switching network **220** to control whether and how power is discharged from the secondary energy storage device **222**.

[0051] In some examples, the controller **202** selectively provides (e.g., by controlling the power switching network **220**) energy stored in the secondary energy storage device **222** to the motor **204** as a motor plugging current in order to brake the motor **204** more quickly during a braking operation (e.g., in response to detecting a motor braking event).

[0052] In some examples, the controller **202** selectively provides the energy stored in the secondary energy storage device **222** to the motor **204** as supplementary power for driving the motor **204**. The supplementary power is provided to the motor **204** in addition to the power provided by the power source **242** during the driving operation of the motor **204**. For example, in some examples, the controller **202** controls the power switching network **220** to provide power from the power source **242** and the supplementary power from the secondary energy storage device **222** at startup of the motor **204**.

[0053] In some examples, the controller **202** provides the supplementary power from the secondary energy storage device **222** in response to determining that the voltage of the power source **242** is below a threshold voltage for driving the motor **204** and/or otherwise powering the tool **100**. In such examples, the controller **202** may provide the supplementary power from the secondary energy storage device **222** as a portion of or all of the total power provided to the motor **204**. For example, the controller **202** may control the power switching network **220** to discharge the secondary energy storage device **222** according to a difference between a power requirement of the tool and an amount of power that can be provided by the power source **242** (e.g., based on a determined voltage state of the power source **242**).

[0054] In some examples, the controller **202** provides the supplementary power from the secondary energy storage device **222** based on a power demand from the motor **204**. For example, in response to determining that a power demand of the motor **204** is greater than a threshold (for example, during different types of driving operations or modes of the tool **100**), the controller **202** may control the power switching network **220** such that power is provided from both the power source **242** and the secondary energy storage device **222** to drive the motor **204**.

[0055] In some examples, the user input module **216** of the tool **100** includes a boost operation input (e.g., a push button, a switch, and/or the like) for triggering a boost mode of the tool **100** (e.g., for driving the motor **204** at a higher power). Responsive to receiving user input via the user input module **216** indicating that a boost operation is requested, the controller **202** controls the power switching network **220** to selectively provide energy stored in the secondary energy storage device **222** as supplementary power for driving the motor **204** during a boost operation. The controller **202** may drive the motor **204** in the boost operation until the controller **202** receives user input deactivating the boost mode, until the energy stored in the energy storage device is **222** is depleted, until a predetermined period of time for performing the boost operation has elapsed, and/or the like.

[0056] In some examples, the controller **202** selectively provides the energy stored in the secondary energy storage device **222** to the power source **242** in response to determining that the motor **204** is in an idle state (e.g., during power down of the tool **100** and/or when the motor **204** is otherwise not being driven) and that a charge state of the power source **242** is below a threshold. For example, in response to determining that the power source **242** is not fully charged, the controller **202** controls the DC-DC power converter **224** to charge the power source **242** using energy stored in the secondary energy storage device **222** by, for example, converting power stored in the secondary energy storage device **222** to a higher voltage level than that of the power source **242** such that a charging current flows from the secondary energy storage device **222** to the power source **242**.

[0057] In some examples, controller **202** controls the DC-DC power converter **224** to regulate power discharged from the secondary energy storage device **222** such that the energy stored in the secondary energy storage device **222** gradually dissipates over time. For example, in response to determining that the motor **204** is in an idle state and that the charge state of the power source **242** is greater than a threshold (e.g., fully charged), the controller **202** controls the power switching network **220** to enable the energy stored in the secondary energy storage device **222** to dissipate. In some examples, the controller **202** enables the energy stored in the secondary energy storage device **222** to dissipate in response to determining that the power source **242** is not connected to the tool **100**.

[0058] In some examples, responsive to determining that a first operating condition is met, the controller **202** may selectively provide energy stored in the secondary energy storage device **222** to the motor **204** (e.g., as supplementary current or as motor plugging current). The first operating condition may include, for example, a braking operation of the tool **100**, an increased power demand of the motor **204**, a startup operation of the motor **204**, a low voltage condition of the power source **242** during driving of the motor **204**, a requested boost operation of the tool **100**, and/or the like.

[0059] Responsive to determining that a second operating condition is met (or the second operating condition is met and first operating condition is not met), the controller **202** may selectively provide energy stored in the secondary energy storage device **222** to the power source **242**. The second operating condition may include, for example, a low voltage condition of the power source **242** (e.g., during an idle state of the motor **204**).

[0060] In some examples, responsive to determining that neither the first nor second operating conditions are met, the controller **202** may enable the energy stored in the secondary energy storage device **222** to dissipate.

[0061] The method **400** illustrates only example embodiments. The blocks described with respect to this method need not all be performed or performed in the same order as described to carry out the method. One of ordinary skill in the art appreciates that the method **400** may be performed with the blocks in any order or by omitting certain blocks altogether.

[0062] Thus, embodiments described herein provide a power tool including a secondary energy storage device. Various features and advantages of the embodiments are set forth in the following aspects:

Claims

1. A power tool comprising: a power input interface electrically connectable to a power source; a motor; a DC-DC power converter electrically connected to the motor; a secondary energy storage device electrically connected to the DC-DC power converter; and a controller electrically connected to the DC-DC power converter, the controller configured to control the DC-DC power converter to provide a current generated by the motor to the secondary energy storage device, the current being generated during motor braking.
2. The power tool of claim 1, wherein the DC-DC power converter is configured to regulate power provided to the secondary energy storage device from the motor during motor braking.
3. The power tool of claim 1, wherein the DC-DC power converter is a bidirectional DC-DC power converter configured to regulate power discharged from the secondary energy storage device.
4. The power tool of claim 1, further comprising: a power switching network electrically connected to the motor and the DC-DC power converter, wherein the controller is configured to control the power switching network to perform motor braking.
5. The power tool of claim 4, wherein the controller is further configured to selectively provide energy stored in the secondary energy storage device to one selected from a group consisting of the motor and the power source.
6. The power tool of claim 5, wherein the controller is configured to selectively provide the energy stored in the secondary energy storage device to the motor as a motor plugging current during a braking operation of the motor.
7. The power tool of claim 5, wherein the controller is configured to selectively provide the energy stored in the secondary energy storage device to the motor as supplementary power for driving the motor.
8. The power tool of claim 7, wherein the controller is configured to provide power from the power source and the supplementary power from the secondary energy storage device to the motor at startup.
9. The power tool of claim 7, wherein the controller is configured to provide the supplementary power in response to determination that a voltage of the power source is below a threshold during a driving operation of the motor.
10. The power tool of claim 7, wherein the controller is configured to provide the supplementary power in response to determination that a power demand of the motor is greater than a threshold.
11. The power tool of claim 7, further comprising: a user input interface; wherein the controller is further configured to receive user input via the user input interface indicating that a boost operation is requested, and selectively provide the energy stored in the secondary energy storage device to the motor as supplementary power for driving the motor in response to determining that the boost operation is requested.
12. The power tool of claim 1, wherein the controller is configured to selectively provide energy stored in the secondary energy storage device to the power source in response to determining that the motor is in an idle state.
13. The power tool of claim 1, wherein the secondary energy storage device includes at least one selected from the group consisting of a capacitor, a supercapacitor, a battery cell, and a power tool

battery pack.

14. The power tool of claim 1, wherein the power source is a removable battery pack.

15. A power tool comprising: a battery pack interface electrically connectable to a battery pack; a motor; a DC-DC power converter electrically connected to the motor; a secondary energy storage device electrically connected to the DC-DC power converter; and a controller electrically connected to the DC-DC power converter, the controller configured to control DC-DC power converter to provide a braking current generated by the motor to the secondary energy storage device during motor braking.

16. The power tool of claim 15, wherein the controller is further configured to responsive to determining that a first operating condition of the power tool is met, selectively provide energy stored in the secondary energy storage device to the motor, responsive to determining that the first operating condition of the power tool is not met and that a second operating condition of the power tool is met, selectively provide the energy stored in the secondary energy storage device to the battery pack, and responsive to determining that the first operating condition and the second operating condition are not met, selectively enable the energy stored in the secondary energy storage device to dissipate.

17. A method for operating a power tool including a power input interface electrically connectable to a power source, a motor, a DC-DC power converter, and a secondary energy storage device, the method comprising: controlling the DC-DC power converter to provide a braking current generated by the motor to the secondary energy storage device during motor braking.

18. The method of claim 17, further comprising: selectively providing energy stored in the secondary energy storage device to one selected from a group consisting of the motor and the power source.

19. The method of claim 18, wherein the energy stored in the secondary energy storage device to the motor is selectively provided to the motor as a motor plugging current during a braking operation of the motor.

20. The method of claim 18, wherein the energy stored in the secondary energy storage device to the motor is selectively provided to the motor as supplementary power for driving the motor.
